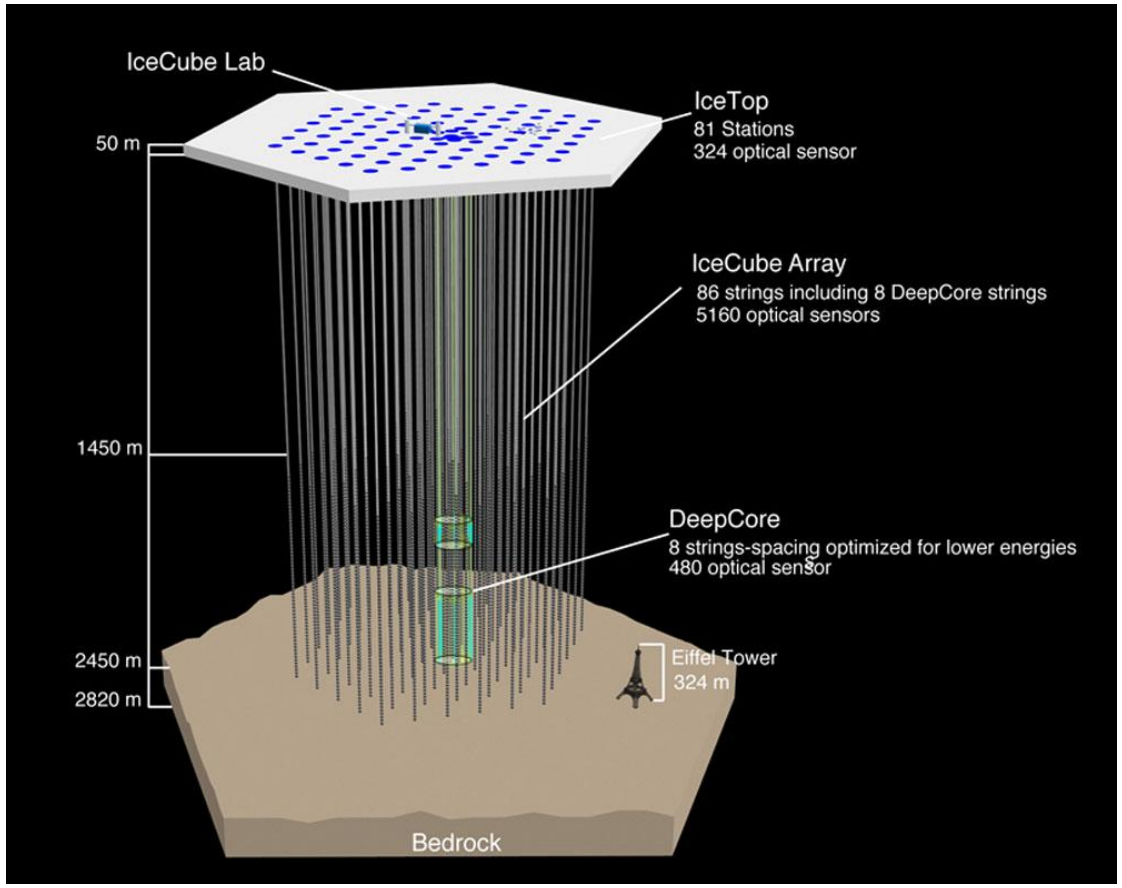
Combining HAWC and IceCube Data Using Rigidity to Map Cosmic Ray Anisotropy

Riya Kore

Paolo Desiati, Juan Carlos Díaz Vélez

Wisconsin IceCube Particle Astrophysics Center

Observatories



IceCube Neutrino Observatory



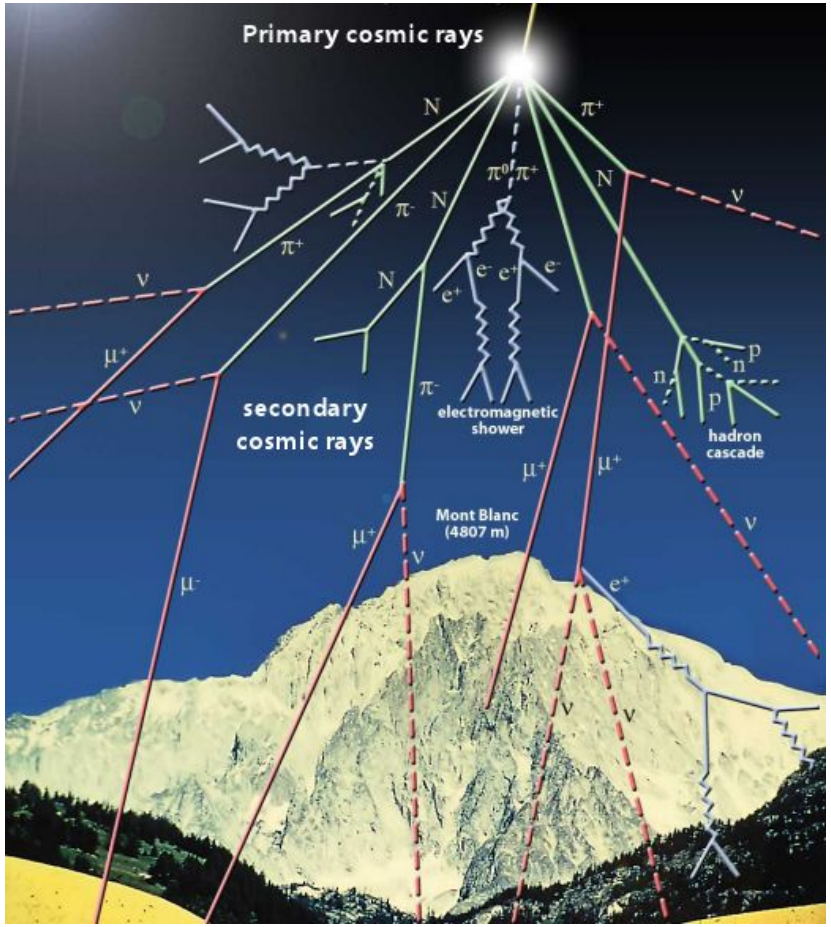
High Altitude Water Cherenkov Observatory

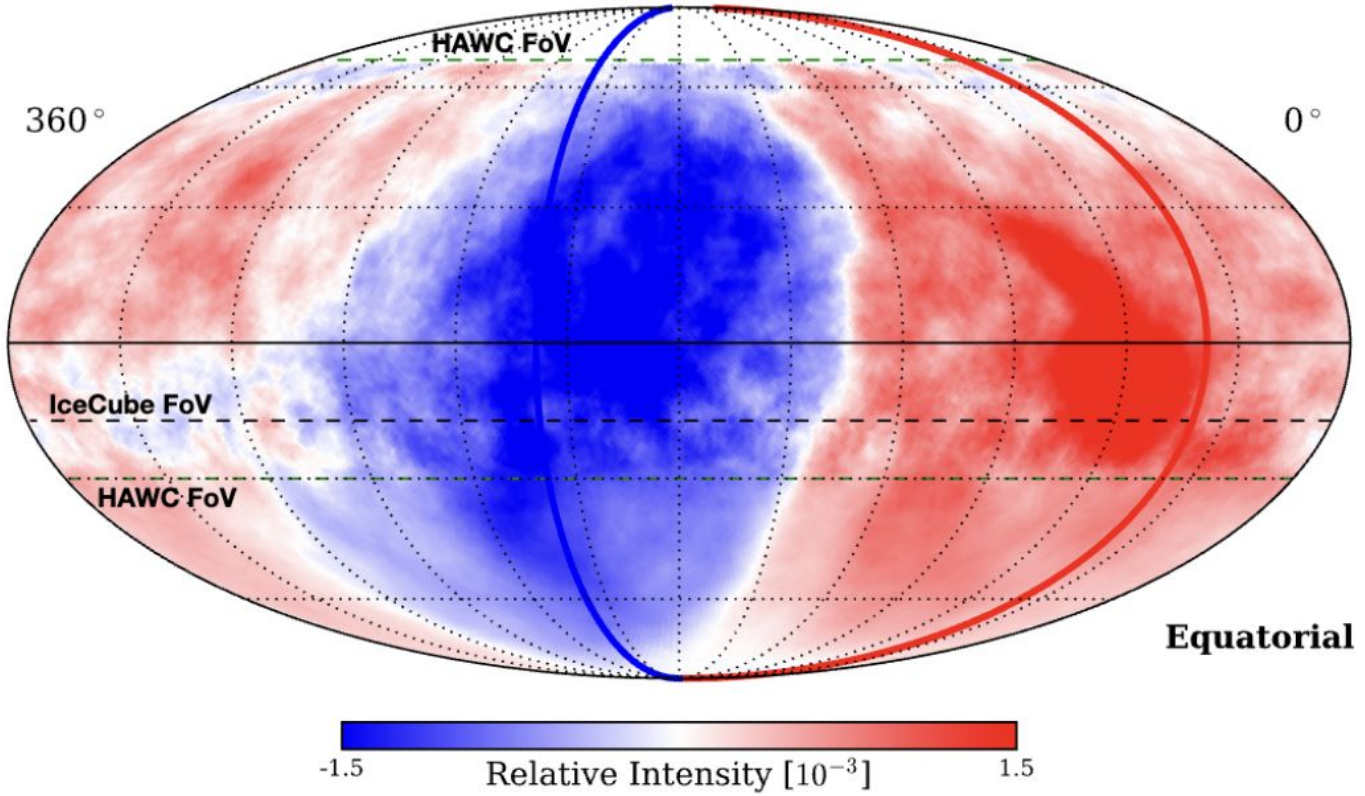
WIPAC Wisconsin IceCube Particle Astrophysics Center

IceCube Neutrino Observatory

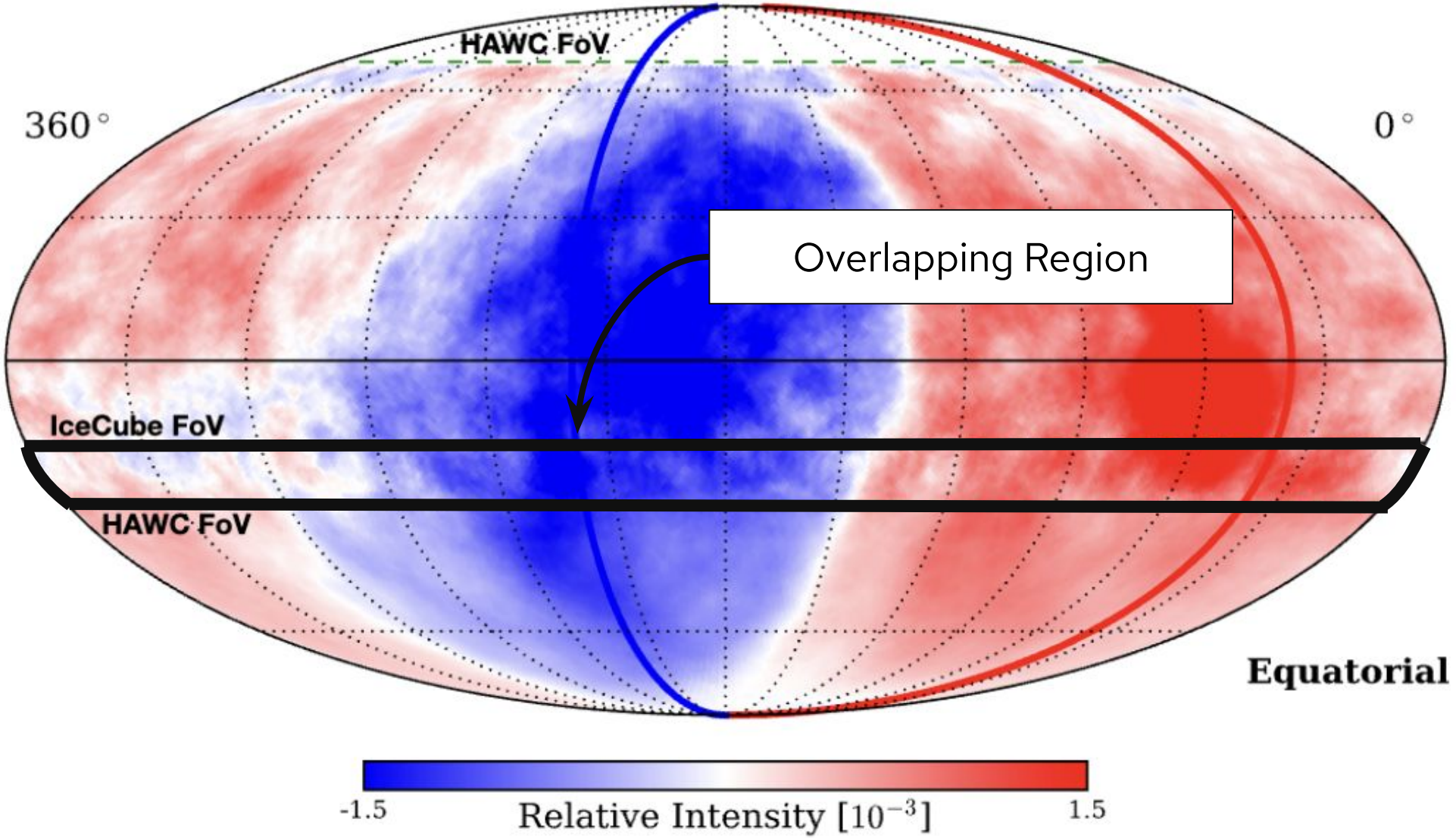
HAWC High Altitude Water Cherenkov Gamma-Ray Observatory

Cosmic Rays and Anisotropy

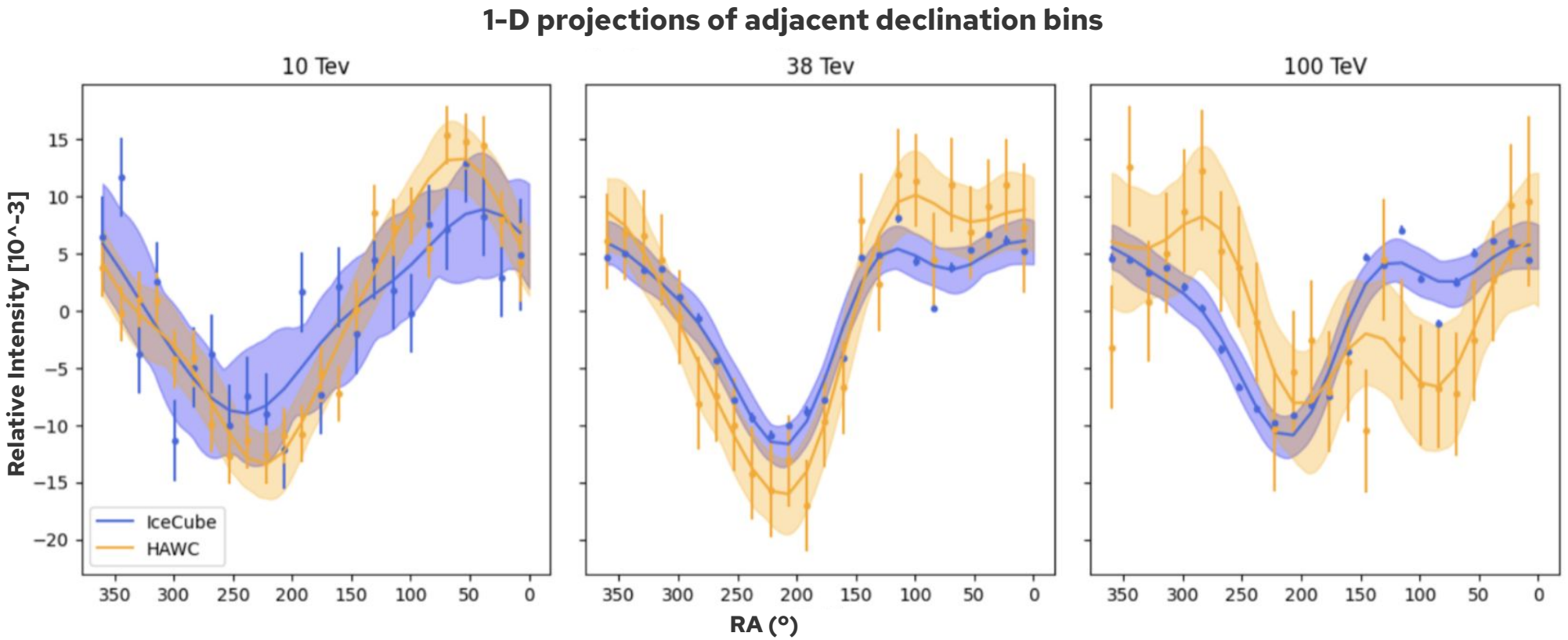




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IceCube Neutrino Observatory
HAWC High Altitude Water Cherenkov Gamma-Ray Observatory


HAWC FoV
360°
0°
Overlapping Region
IceCube FoV
HAWC FoV
Equatorial
-1.5
Relative Intensity [10^{-3}]
1.5

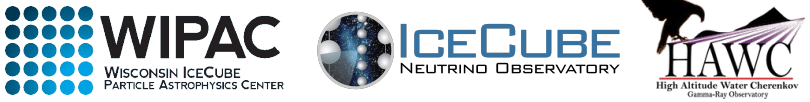
Previous Work on Overlapping Region



- Old analysis
- Matching done in energies but not compatible

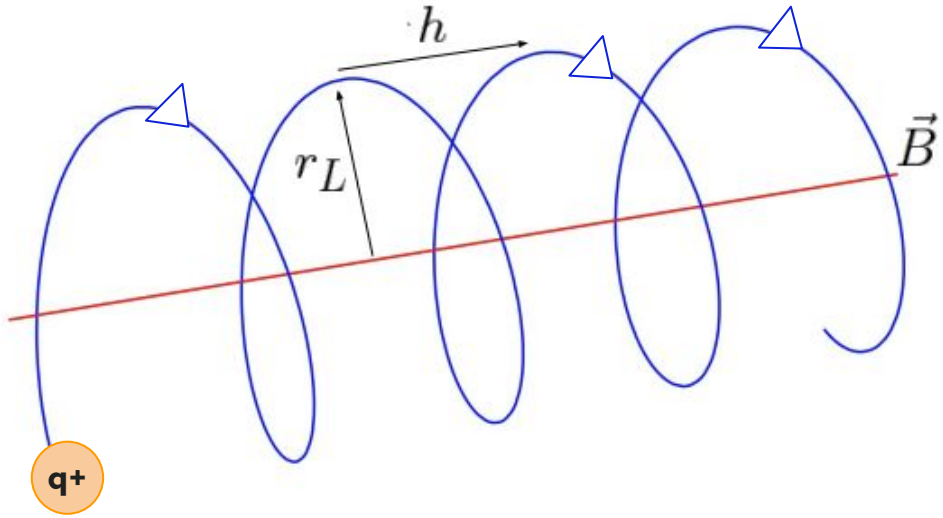
So, we came up with a new approach!





Magnetic Rigidity

- Path because of Lorentz Force
  $\vec{F} = q(\vec{v} \times \vec{B})$
- Larmor Radius (r_L) is the Rigidity of the charged particle
- $\text{Rigidity} = mv/q\vec{B}$ for non-relativistic particles
- For cosmic rays: $\text{Rigidity} = E/Z$



Hypothesis

- Cosmic Ray Anisotropy is a function of Rigidity.
- HAWC and IceCube have different sensitivities to Cosmic Ray mass and charge, and that is why we need to scale with Rigidity.

My research is about implementing this approach, testing it on the simulation data and then applying it to the real data and generating the all sky maps!

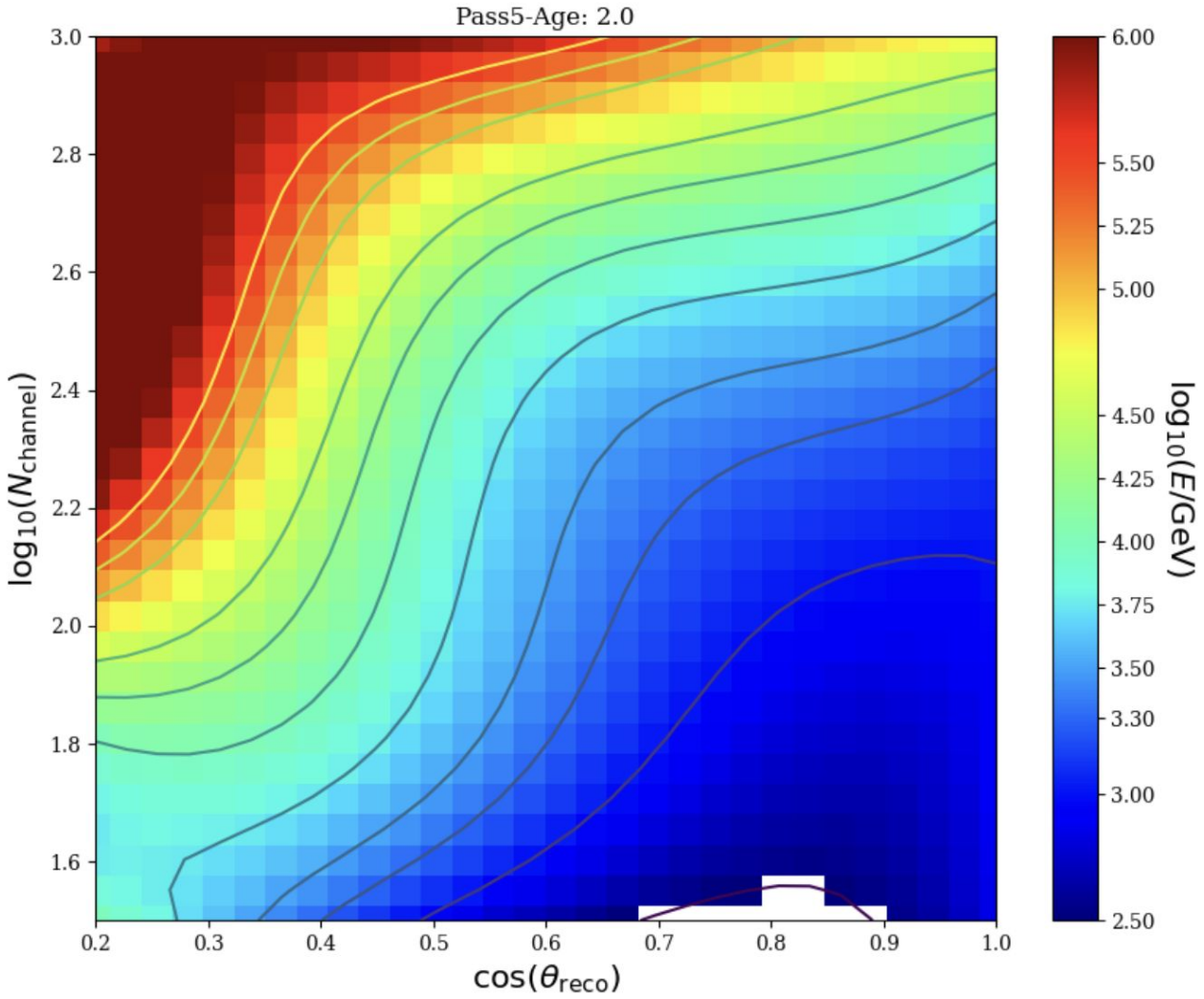
Methodology

1. Binned the per event Rigidity for HAWC in terms of Energy.

2. Performed the Kolmogorov-Smirnov Test (KS Test) between HAWC bins and each window of IceCube energy range to compare Rigidities.

3. Got the most compatible energy ranges for HAWC and IceCube based on the KS Statistic.

4. Made the All Sky Maps for those energy ranges using the real data



- Selection criteria for events at different energies to get a statistical energy range.
- Binned HAWC Rigidities using this selection.

Methodology

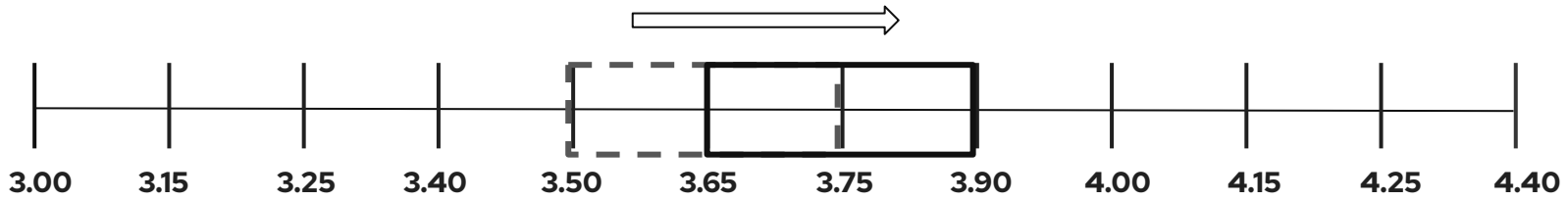
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1. Implemented a sliding window approach on IceCube energies.

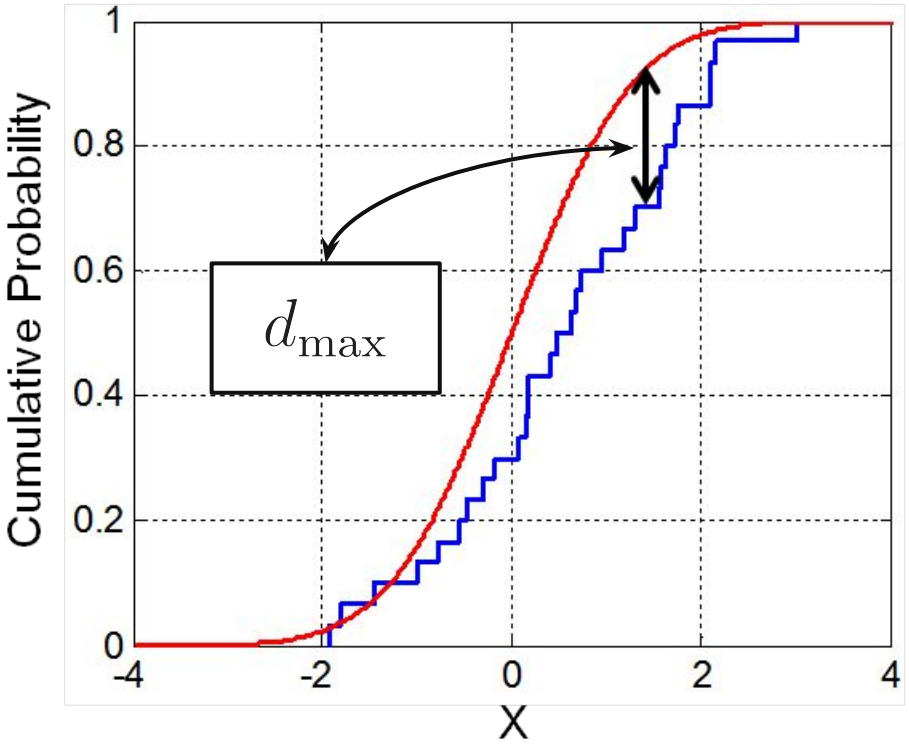


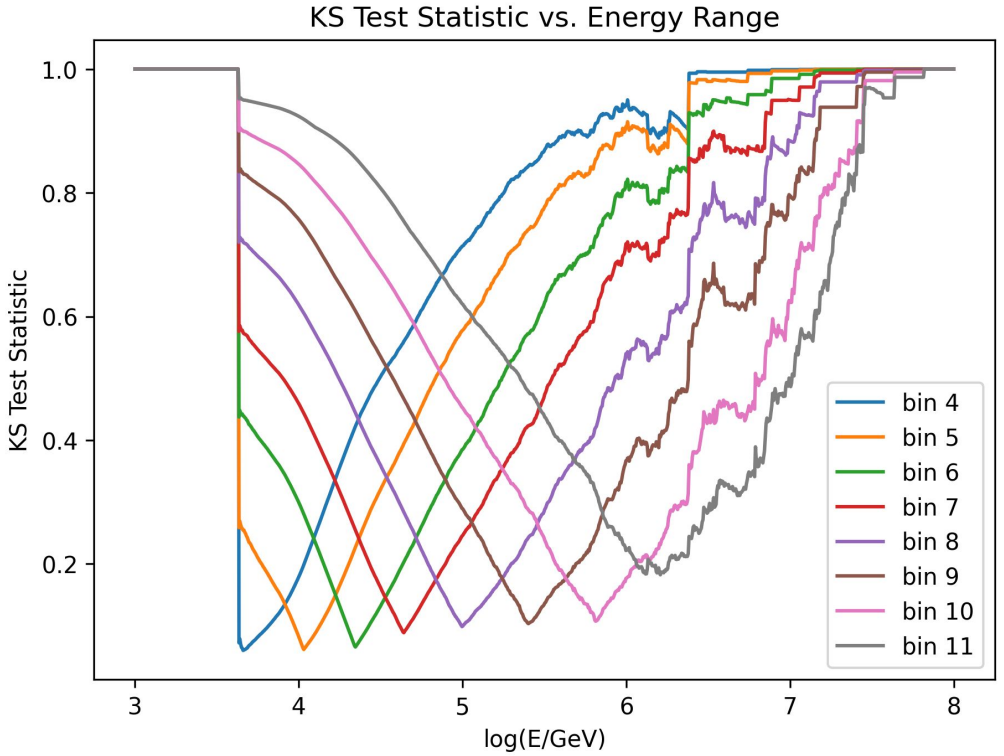
an example of the sliding window approach

2. Calculated the Weighted Cumulative Distribution Functions (CDFs) for HAWC and IceCube Rigidities and found the maximum distance between them.

$$F(x) = \frac{1}{W} \sum_{i=1}^{n} w_i \{d_i \leq x\}$$

3. Found the minimum of those distances. (KS Test)





This gave me the IceCube energy windows most compatible with HAWC in terms of rigidity.

Methodology

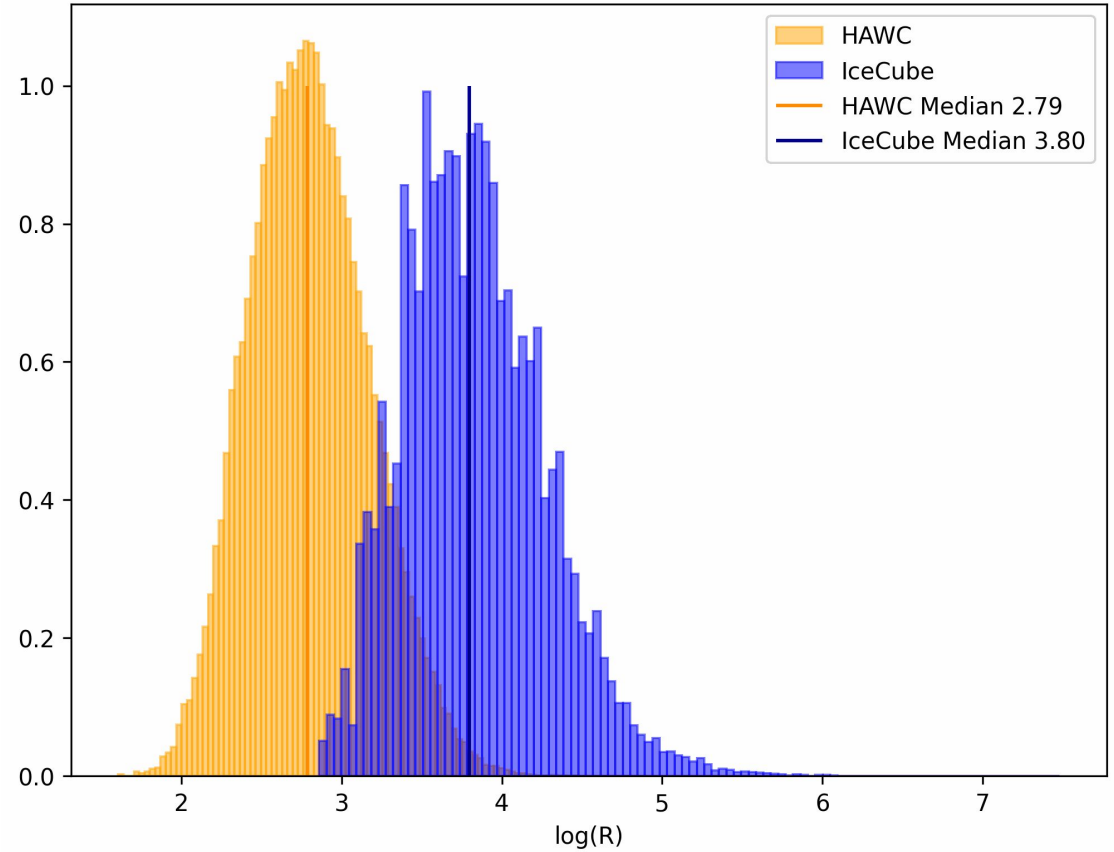
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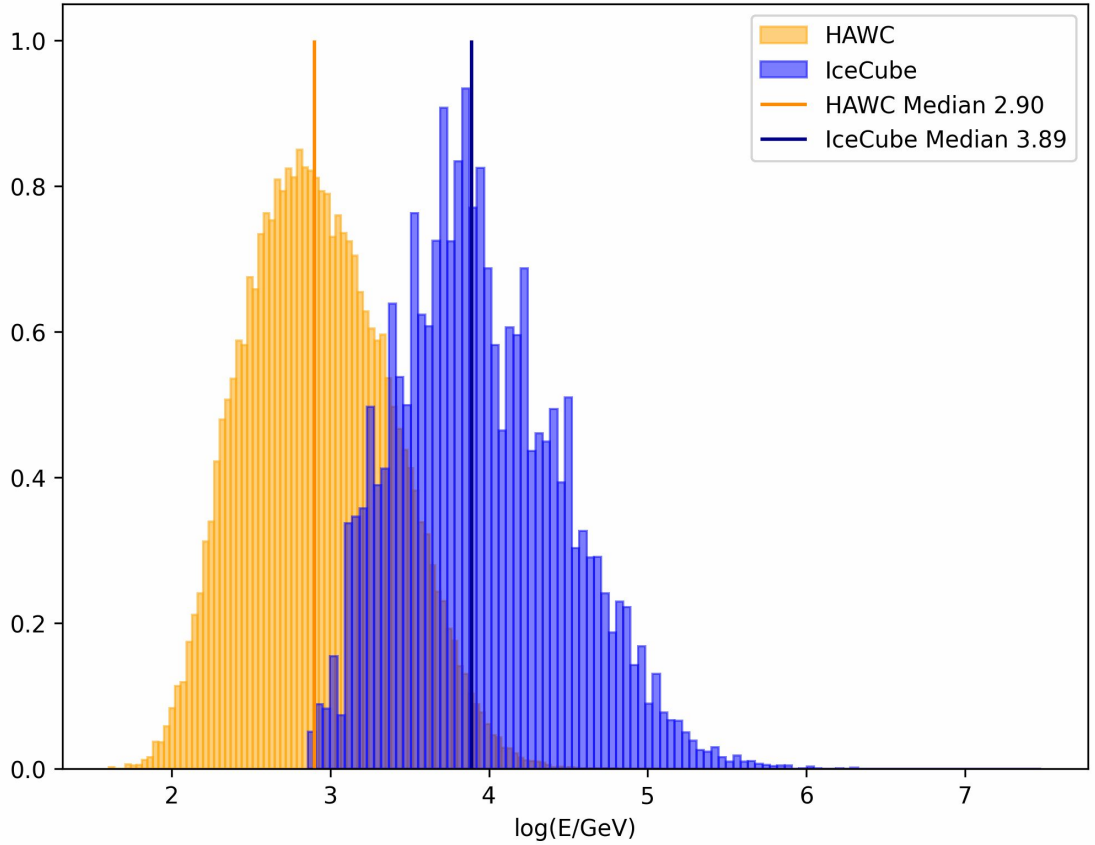
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4. Made the All Sky Maps for those energy ranges using the real data

Results from the simulation analysis!



Rigidity



Energy

Methodology

1. Binned the per event Rigidity for HAWC in terms of Energy.

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1. Calculate the Relative Intensity.

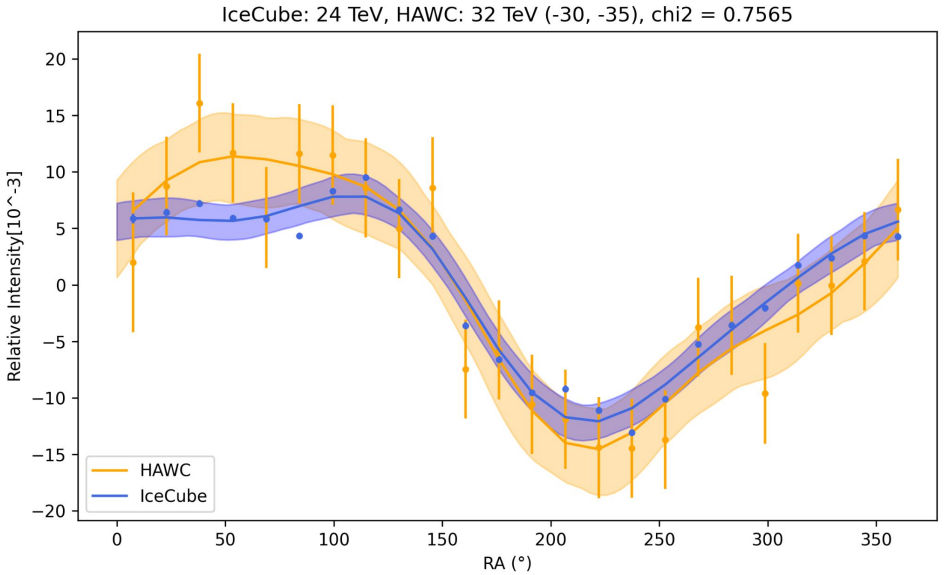
$$\text{Relative Intensity} = \frac{\text{Data}}{\text{Background}} - 1$$

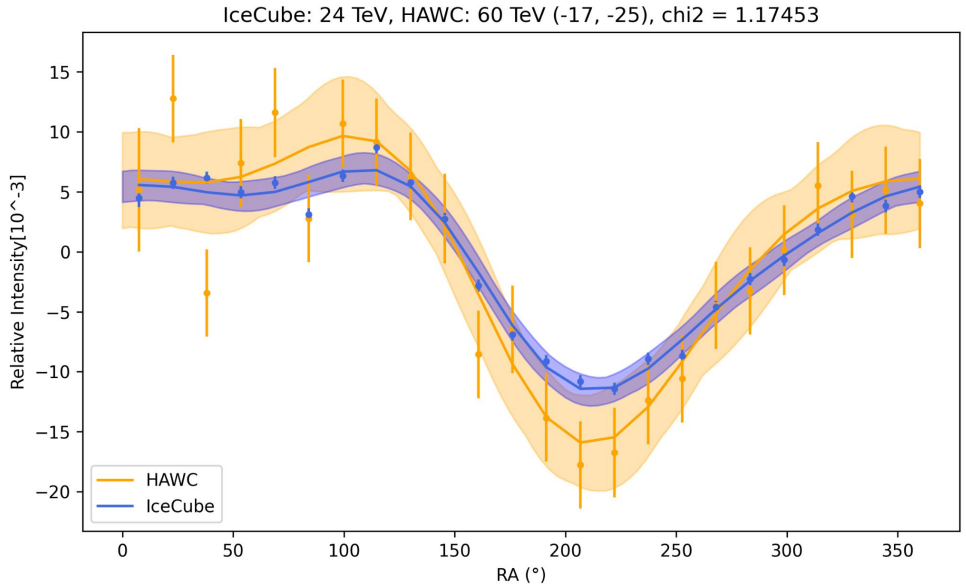
2. Chi-Squared Test: For a given HAWC map, iterate through the IceCube maps and find the energy bin with the lowest χ^2

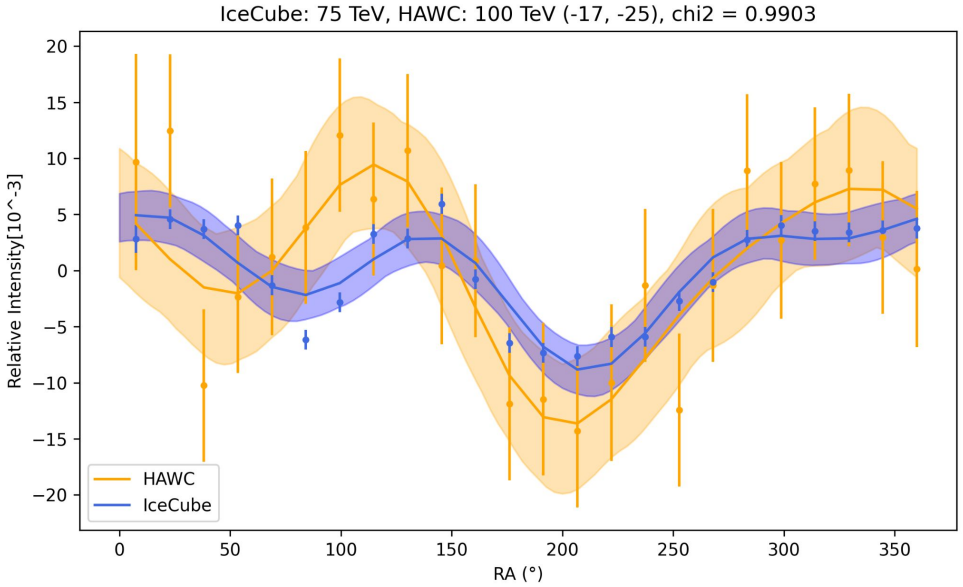
$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

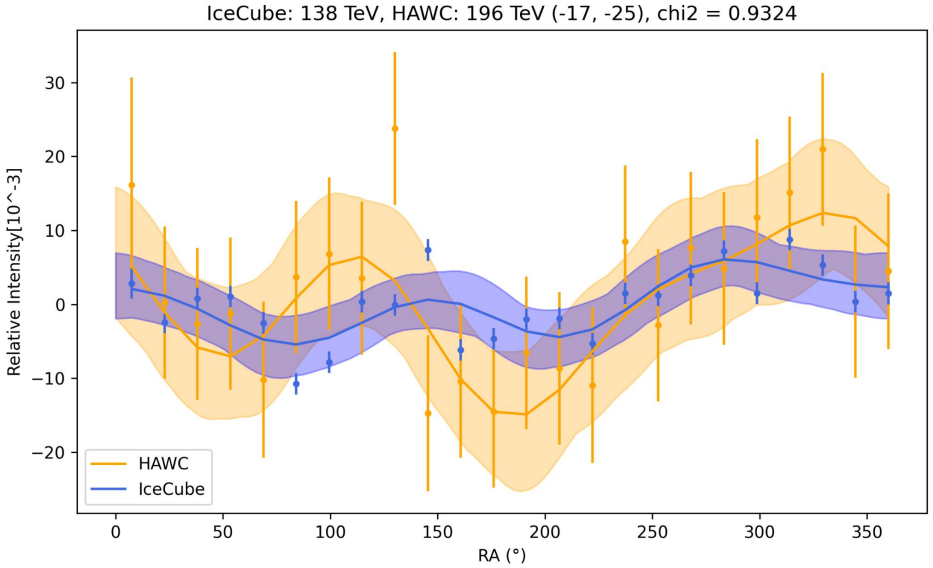
Results!

These are the All Sky Maps of the real data.



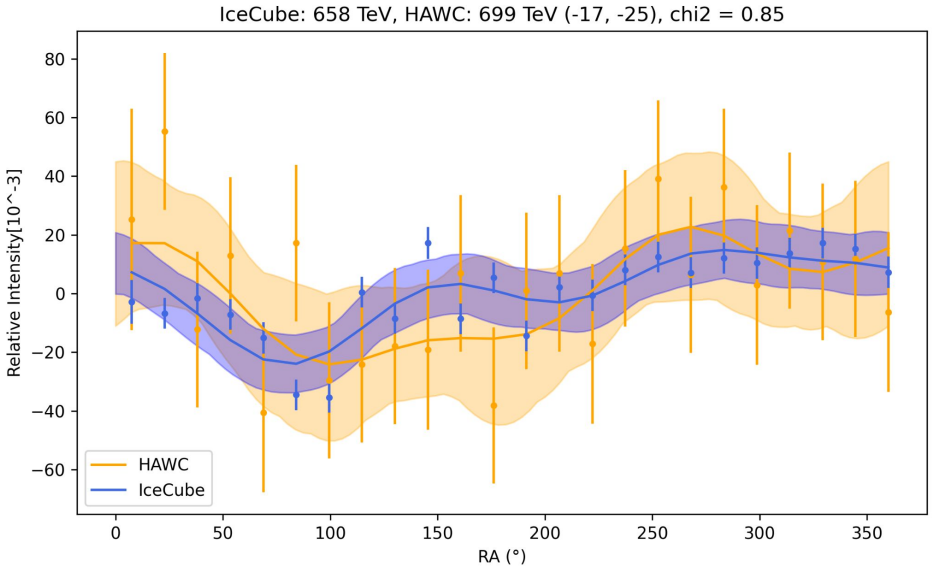


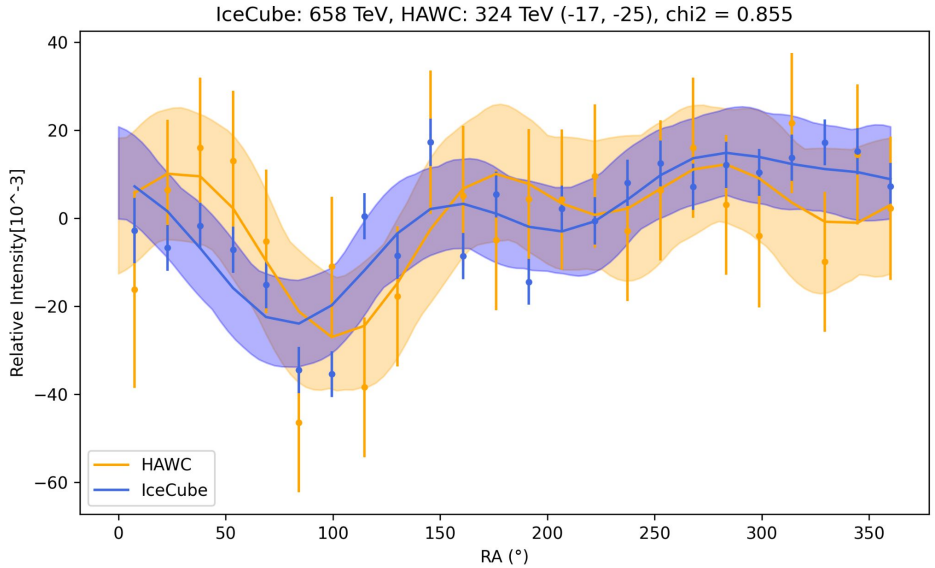




From Monte Carlo analysis:

HAWC 35.65 TeV -> IceCube 14.5 TeV
HAWC 69.9 TeV -> IceCube 30.54 TeV
HAWC 108.5 TeV -> IceCube 60.635 TeV
HAWC 208 TeV -> IceCube 137.3 TeV
HAWC 377 TeV -> IceCube 349.9 TeV
HAWC 739 TeV -> IceCube 900 TeV





Conclusions

- The best matching energy bins are consistent with the Rigidity assumption.
- So, IceCube and HAWC maps for various energy ranges match better in Rigidity than they do in Energy.
- This confirms our theory that Cosmic Ray Anisotropy is a magnetic process.

Future Work

- Get more accurate real IceCube data using the compatible energy ranges from the Monte Carlo simulation analysis.
- Perform the Chi-squared test and make even better All Sky Maps.
- Publish the results!



Thank You!

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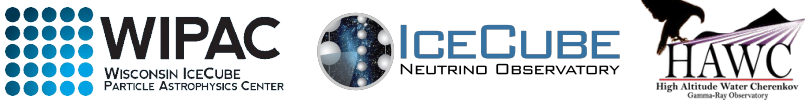


IceCube
Neutrino Observatory



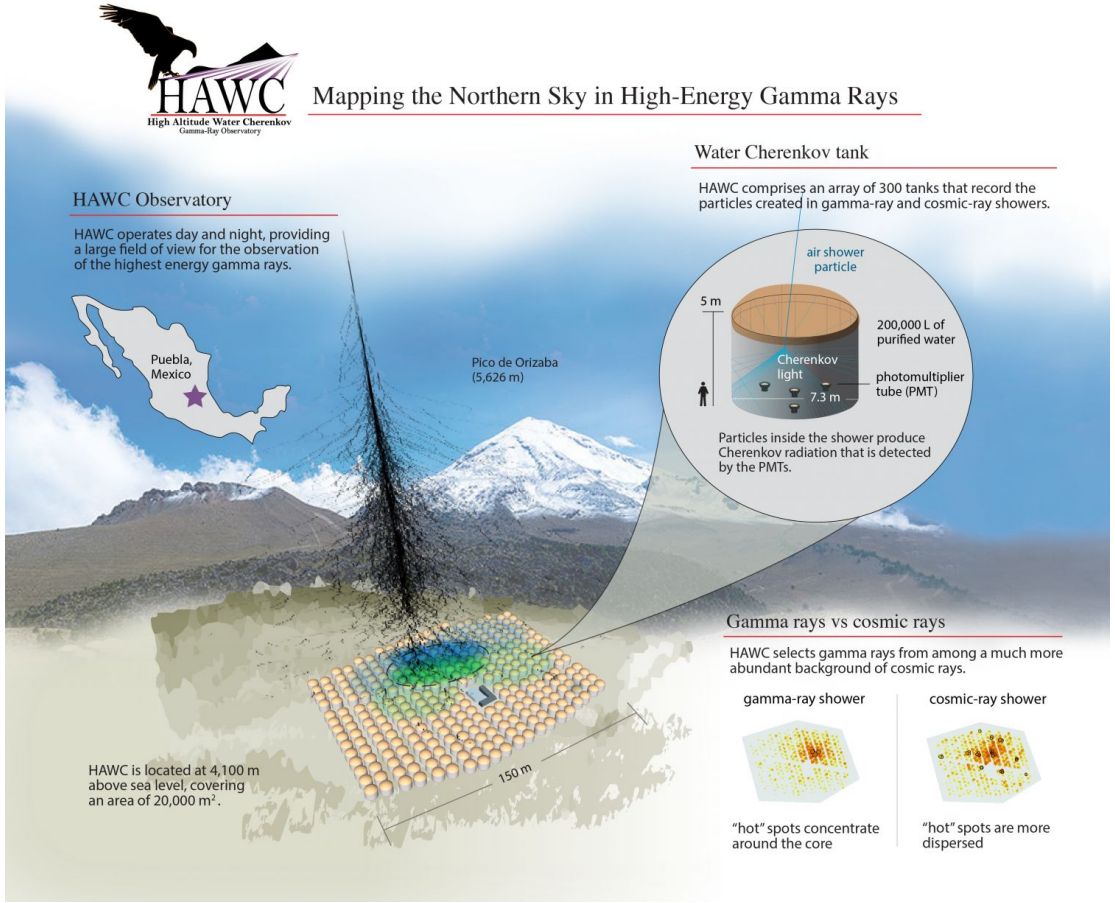
Backup Slides





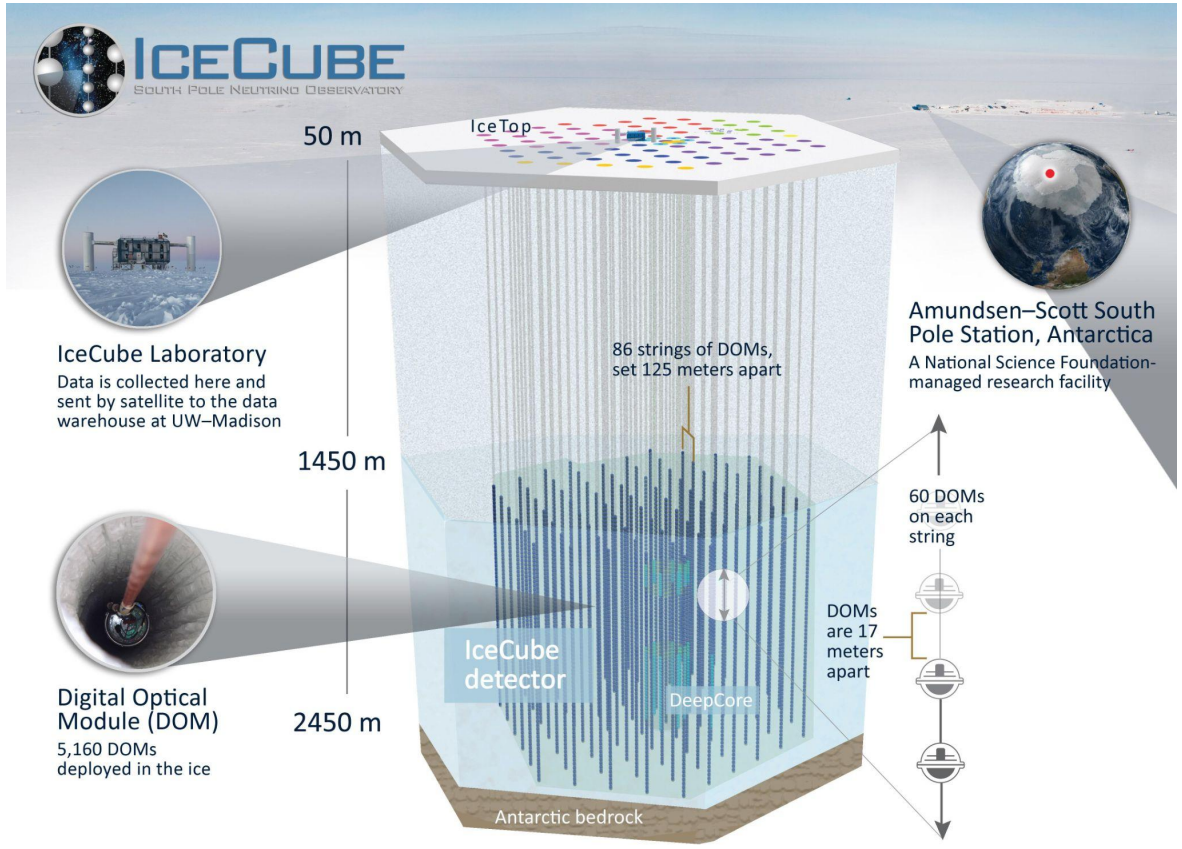
HAWC Observatory

- HAWC is located in Mexico
- Sensitive to energy ranges from 100 GeV to 100 TeV

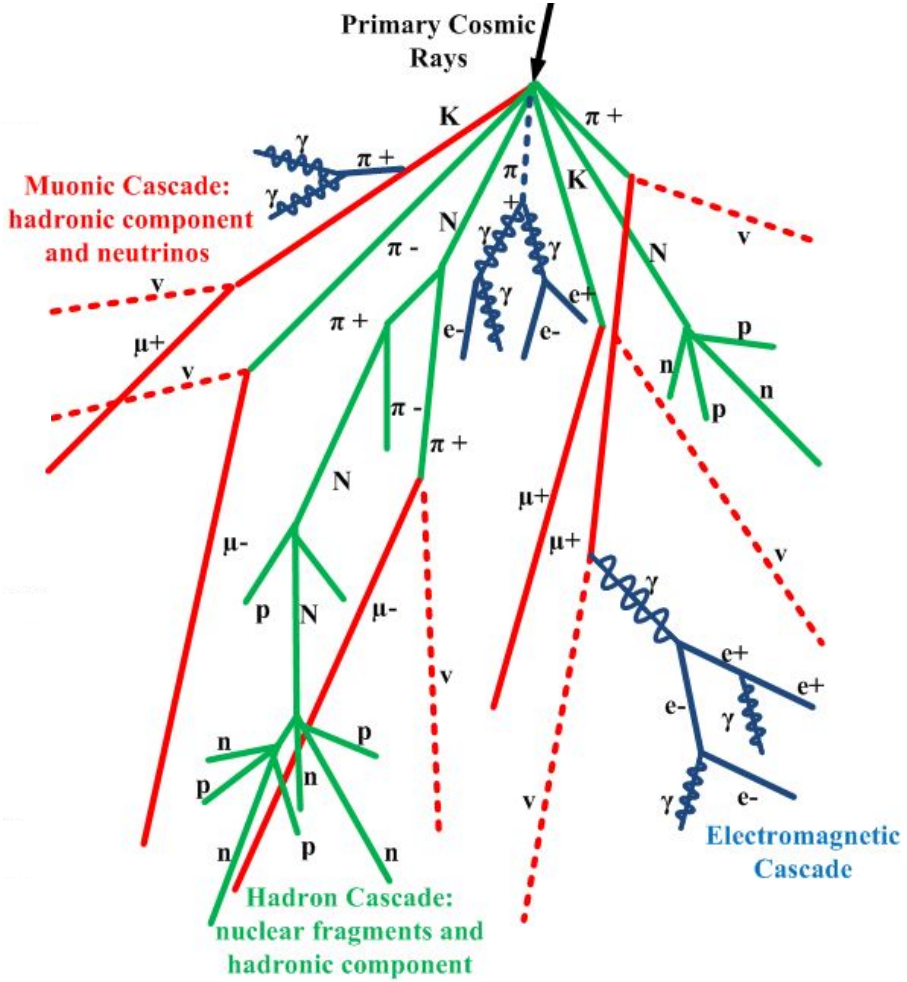


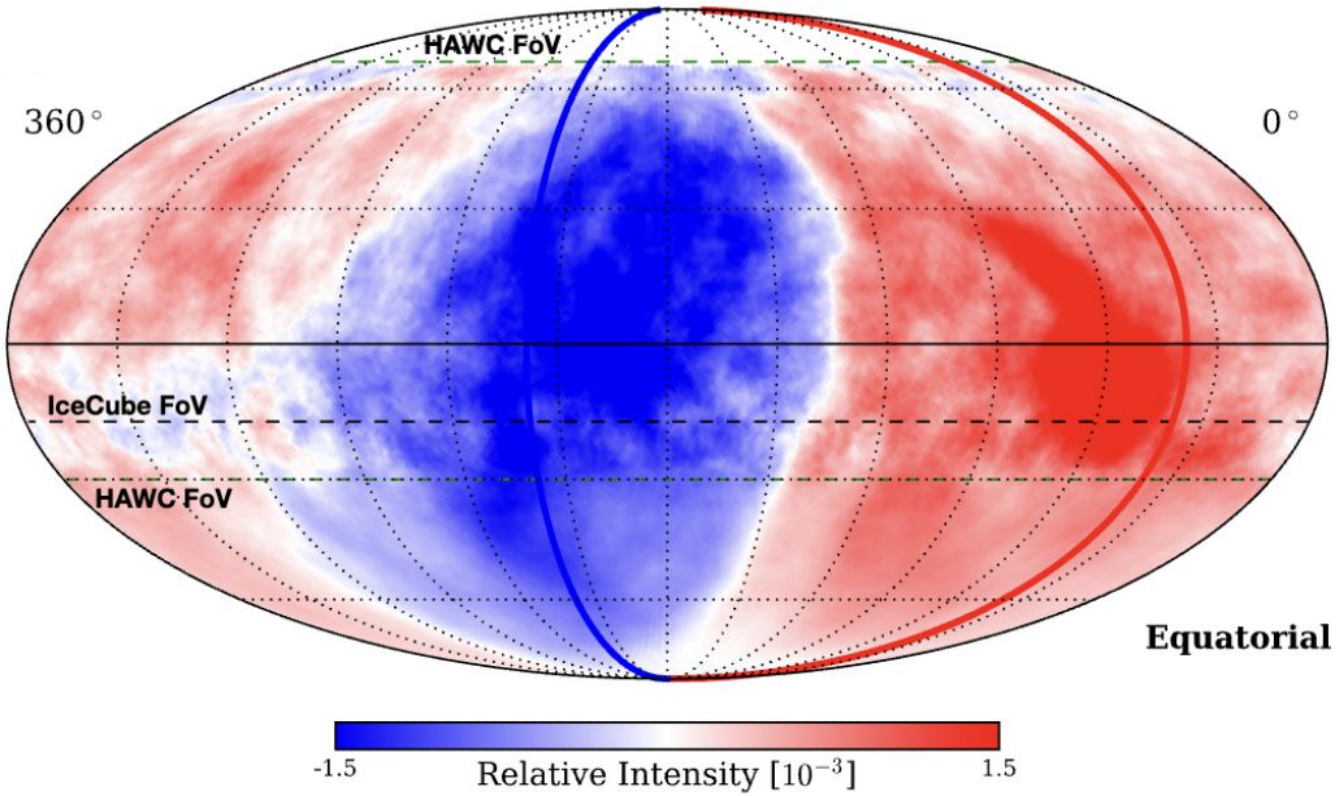
IceCube Neutrino Observatory

- IceCube is located at the South Pole
- Sensitive to energy ranges from 100 GeV to several PeV

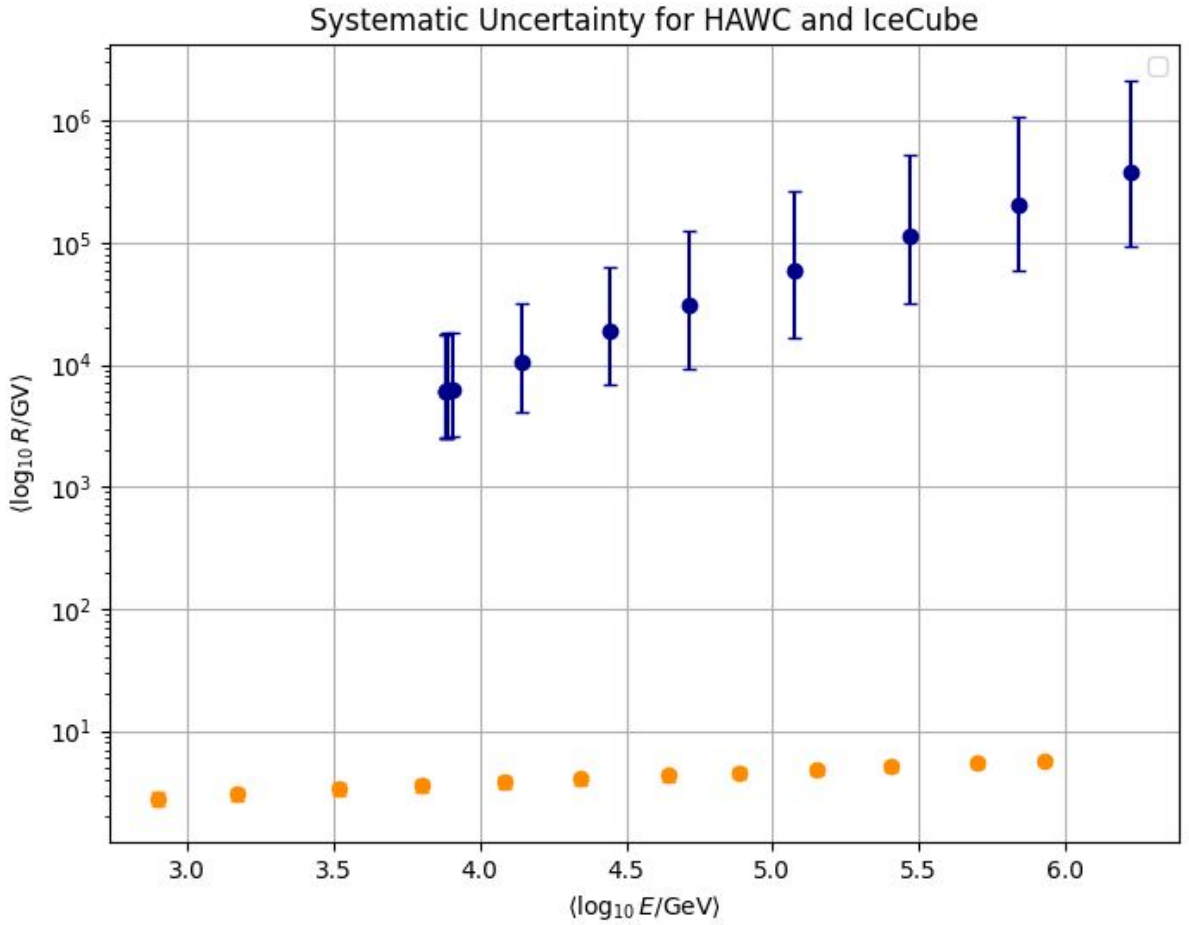


Cosmic Rays and Anisotropy





Systematic Uncertainty



Composition Assumptions

